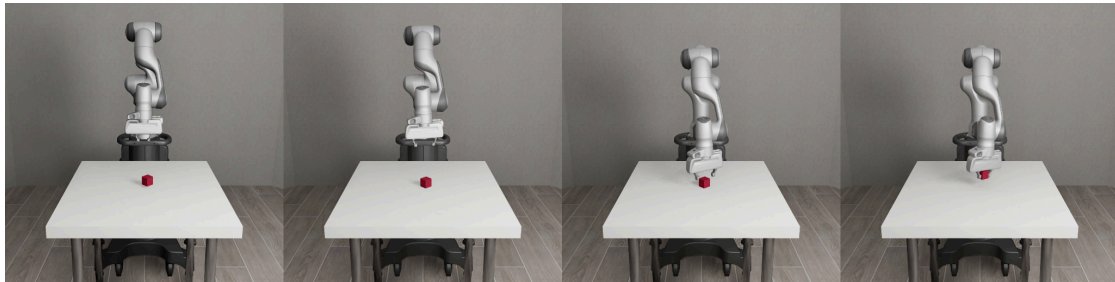


COM SCI 188: Introduction to Robotics (spring 2025)

Coding Assignment 2: Camera Calibration

Deadline: 5/2 11:59pm

In the last coding assignment, you had access to the ground-truth simulation state for the block. In practice, you can only observe the environment using cameras!



(Lift Task)

In this assignment, you will implement a calibration routine to compute the transformation matrix from camera frame to robot frame and perform the Lift task using only image observation (alongside proprioception data).

Steps:

1. **Detect red blobs in rgb image (detect_red_blob):** implement a function that detects the largest red blob in the scene and returns the center pixel
2. **Convert pixel to 3D point in camera frame (pixel_to_camera3d):** given camera intrinsics and depth map, compute the 3D coordinate of a 2D pixel
3. **Implement a calibration routine (run_hand_eye_calibration):** design waypoints for the robot and collect paired waypoints in camera frame and robot frame
4. **Implement Kabsch algorithm to find the transformation matrix (solve_for_rigid_transformation):** given two sets of paired waypoints, find the rigid-body transformation matrix that best transform one into the other
5. **Find the calibration offset (get_calibration_offset):** there is often an offset introduced during calibration and needs to be accounted for through manual fine-tuning

Code Submission:

- Please use the provided skeleton code.
- Submit camera_utils.py and T_CR.npy through gradescope. Do not rename the functions!

Resources:

- [Robosuite Documentation](#)